SASTRA
ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act,1956)
THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

DOCUMENTATION

Cloud Cover Estimation Using Multiple Linear Regression: A Case Study on Chennai Weather Data

Documented by :

Venkateswaran M

(327150024)

I M.Sc Data Science,

SASTRA,Chennai Campus.

ABSTRACT:

Cloud cover plays a crucial role in weather forecasting, solar radiation analysis, and climate modeling. Accurate estimation of cloud cover helps in improving solar energy forecasts, aviation planning, and atmospheric research.

In this project, we developed a data-driven approach to predict **cloud cover percentage** using multiple meteorological parameters such as temperature, humidity, visibility, pressure, wind, and solar radiation.

The study utilizes an **Exploratory Data Analysis (EDA)** approach to identify significant factors influencing cloud cover, followed by a **multiple linear regression model** built using NumPy's least squares method. Feature engineering, interaction terms, and normalization techniques were employed to enhance prediction accuracy.

The final model achieved an **R^2 score of around 0.70**, indicating strong linear dependence between selected predictors and cloud cover. The project further includes **residual analysis, error distribution visualization, feature importance ranking**, and **symbolic differentiation** to interpret variable sensitivity.

TABLE OF CONTENTS

1.INTRODUCTION

Cloud cover plays a vital role in weather forecasting, climate modeling, and environmental studies. It affects temperature regulation, solar radiation reaching the Earth’s surface, and even rainfall patterns. Accurate estimation of cloud cover helps meteorologists improve short-term weather forecasts and supports long-term climate monitoring. However, traditional methods for estimating cloud cover often rely on satellite imagery and complex atmospheric simulations, which can be expensive, computationally demanding, and sometimes inaccessible for localized studies.

To address this challenge, data-driven approaches offer a simpler and more efficient alternative. By analyzing relationships between easily measurable surface-level parameters—such as temperature, humidity, air pressure, wind speed, and precipitation—it is possible to estimate cloud cover with reasonable accuracy. This project focuses on developing a regression-based model that predicts cloud cover over Chennai using historical weather data.

The main goal is to create a computationally efficient model using basic statistical and mathematical concepts, without depending on external machine learning libraries. The study involves exploring the data through descriptive statistics and visualizations, identifying key factors influencing cloud formation, and building a linear regression model from scratch using NumPy. The performance of the model will be evaluated using standard regression metrics, and the insights gained can be used to better understand local atmospheric behavior. This approach not only simplifies the modeling process but also demonstrates how surface weather data can be used effectively for predicting cloud cover in a resource-efficient manner.

2. OBJECTIVES

1. Perform Exploratory Data Analysis (EDA)

To explore and understand the dataset by identifying patterns, correlations, and trends among meteorological variables that influence cloud cover.

2. Develop a Multiple Linear Regression Model

To construct a regression model using the **least squares fitting method** implemented in **NumPy**, predicting cloud cover based on key surface-level weather parameters.

3. Introduce Interaction Terms

To incorporate interaction effects between variables, capturing potential **nonlinear relationships** that improve model performance and interpretability.

4. Evaluate Model Performance

To assess the accuracy and reliability of the model using statistical metrics such as the **Coefficient of Determination (R^2)** and **Root Mean Squared Error (RMSE)**.

5. Perform Error Analysis

To analyze residuals and identify sources of prediction errors, highlighting model strengths and limitations in representing real-world cloud behavior.

6. Interpret Variable Sensitivities

To use **symbolic partial derivatives** for understanding how changes in each input variable affect predicted cloud cover, providing deeper insight into the model's responsiveness.

7. Visualize and Communicate Results

To present findings through clear and data-driven **charts, plots, and visual summaries**, making the relationships and model outcomes easily interpretable.

3. DATASET DESCRIPTION

Dataset: Chennai Weather Dataset
Source: Local weather archive / open-source meteorological data.
Size: Several hundred daily weather observations.

Column Name	Description
temp, tempmax, tempmin	Temperature parameters (°C)
humidity	Relative humidity (%)
visibility	Average visibility (km)
Sea level pressure	Atmospheric pressure (hPa)
windspeed, wind direction	Wind characteristics
Solar radiation, solar energy	Solar radiation and energy
Uv index	UV exposure index
dew	Dew point temperature
Cloud cover	Target variable (percentage)

Derived / Engineered Features:

- temp_range = tempmax - tempmin
- solar_ratio = solarradiation / max(solarradiation)
- dew_diff = temp - dew
- Interaction terms:
  - i1 = t × h
  - i2 = h × vis
  - i3 = t × vis
  - i4 = t × h × vis

4. METHODOLOGY

The methodology followed in this project consists of four major stages: **data preprocessing**, **model development**, **performance evaluation**, and **sensitivity analysis**. Each stage was carefully designed to ensure the reliability and interpretability of the cloud cover prediction model.

4.1 Data Preprocessing

The dataset used in this study consists of historical meteorological observations from Chennai, including variables such as temperature, humidity, air pressure, wind speed, visibility, and precipitation. To ensure data quality and consistency, several preprocessing steps were carried out:

- **Feature Engineering:**
  Additional derived features were introduced to capture potential dependencies not directly visible in the raw data. Examples include dew point estimation, humidity ratio, or temperature difference between consecutive days.
- **Normalization:**
  To ensure that all features contribute equally to the regression model, variables were normalized using **Z-score normalization**.
- **Tools :** Python Libraries: Pandas, Seaborn, Numpy, Matplotlib, Sympy

4.2 Model Development

A **Multiple Linear Regression (MLR)** model was developed manually using **NumPy's lstsq()** function, which performs **least squares fitting** to determine optimal regression coefficients.

The dataset was divided into two subsets:

- **Training Set (80%)** – used to estimate model parameters.
- **Testing Set (20%)** – used to validate and evaluate model performance.

To capture subtle nonlinear relationships, **interaction terms** between key features (e.g., temperature × humidity) were introduced, enhancing the model's ability to represent complex dependencies in atmospheric behavior.

4.3 Performance Metrics

The model's performance was assessed using multiple evaluation criteria:

- **R^2 Score (Coefficient of Determination)**
- **Root Mean Square Error (RMSE)**
- **Residual Analysis:**
  Residuals (differences between predicted and actual values) were analyzed to detect systematic biases and check the assumption of normally distributed errors.

4.4 Sensitivity Analysis

To interpret how different meteorological variables influence cloud cover, a **sensitivity analysis** was conducted using **symbolic computation** with the **Sympy** library. Partial derivatives of the regression equation were computed with respect to key variables such as temperature ((T)) and humidity ((H))

These derivatives quantify how small changes in each variable affect the predicted cloud cover, providing insights into the model's responsiveness and helping identify the most influential factors.

Visualizations of sensitivity trends were also generated to make the relationships between cloud cover and independent variables more interpretable.

---

5. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was conducted to gain insights into the structure of the dataset, identify important relationships among variables, and detect potential patterns that influence cloud cover formation. The analysis focused on understanding how different meteorological factors—such as temperature, humidity, wind speed, and visibility—interact and contribute to cloud development over Chennai.
The following key analyses were performed:

5.1 Correlation Heatmap

A **correlation heatmap** was generated to quantify the linear relationships between variables.
The results revealed a **strong positive correlation** between **humidity and cloud cover**, indicating that higher humidity levels are generally associated with increased cloud formation.
In contrast, **visibility** showed a **moderate negative correlation** with cloud cover, suggesting that lower visibility conditions often coincide with higher cloud density.

5.2 **Feature Importance Plot**

The linear regression coefficients were visualized using a horizontal bar chart to interpret the **relative importance of features**.

Features with smaller coefficients contributed less to model performance, providing a clear understanding of variable influence.

5.3 **Error Distribution**

A histogram of residuals (prediction errors) was plotted to evaluate the model's bias and variance.

The **error distribution was centered around zero** with a **low spread**, indicating that the model is largely unbiased and consistent.
This confirmed that the assumptions of linear regression—particularly normality and homoscedasticity—were reasonably satisfied.

5.4 **Actual vs. Predicted Cloud Cover**

A scatter plot comparing **actual** and **predicted** cloud cover values was created, with a reference diagonal line representing perfect prediction.

Most data points clustered closely around the line, showing a **strong agreement between predicted and observed values**.

This visualization supported the model's **R^2 score** and confirmed good generalization on test data.

5.5 **Temporal Analysis of Cloud Cover and Humidity**

A time-series plot was used to visualize **temporal variations** in cloud cover and humidity across the observation period.

It revealed that **periods of high humidity consistently corresponded to increased cloud cover**, reinforcing the earlier correlation findings.
This analysis also highlighted **seasonal trends and atmospheric variability** specific to the Chennai region.

Through this comprehensive EDA, key influencing variables were identified, guiding the selection of predictors for the regression model. The analysis also revealed the importance of considering both linear and interaction effects when modeling cloud cover dynamics.

6. RESULTS AND EVALUATION

The developed multiple linear regression model was evaluated using standard statistical metrics and visualization-based diagnostic tools. These evaluations helped assess the model's predictive accuracy, stability, and ability to capture the underlying relationships between meteorological variables and cloud cover.

6.1 Model Performance Metrics

Metric	Value	Interpretation
R^2 Score	**0.70**	The model explains approximately **70% of the variance** in observed cloud cover values, indicating a strong overall fit.
RMSE	**≈ 10.5**	On average, the model's predictions deviate about 10.5 percentage points from actual observations.
Mean Error	**≈ 0**	The near-zero mean error suggests that the model is **unbiased**, with no consistent overestimation or underestimation trend.
Standard Deviation of Error	**Low variance**	Indicates that residuals are **consistent and stable**, supporting the model's reliability.

6.2 Interpretation of Results

- The regression model achieved a **reasonable predictive capability**, explaining around **70%** of cloud cover variability based on surface-level parameters.
- **Humidity** emerged as the **most dominant predictor**, confirming its strong influence on cloud formation as observed in the EDA phase.
- The inclusion of **interaction terms**—such as temperature × humidity (t*h) and temperature × humidity × visibility (t*h*vis)—**enhanced model stability** and improved generalization on unseen data.
- The **relationship between cloud cover and temperature** was found to be **inverse**, meaning higher temperatures generally correspond to reduced cloud density.

6.3 Residual Analysis

A residual distribution plot was used to evaluate the difference between predicted and actual cloud cover values.

- The residuals were **symmetrically distributed around zero**, indicating that the model effectively captured the main trends without systematic bias.
- No clear pattern was observed in the residual scatter, suggesting that **linear regression was an appropriate choice** for the dataset.
- Slight deviations at extreme values may be attributed to nonlinear atmospheric effects not fully captured by the linear model.

6.4 Feature Importance Analysis

A **feature importance plot** was generated to assess the relative influence of each predictor on cloud cover.

- **Humidity** and **temperature** were identified as the **top contributors**, aligning with physical meteorological expectations.
- **Visibility** and **pressure** had moderate influence, while **wind speed** contributed relatively less.
- The results confirm that **moisture-related parameters** play the most significant role in determining cloud cover levels in the Chennai region.

7. SENSITIVITY ANALYSIS

To better understand how variations in individual meteorological parameters influence predicted cloud cover, a **sensitivity analysis** was conducted using **symbolic differentiation**. This approach quantifies the effect of small changes in input variables on the model's output, providing valuable interpretability for the regression model.

7.1 Symbolic Derivatives

Using the regression equation derived in the model, **partial derivatives** were computed symbolically with respect to key predictors — **temperature (T)** and **humidity (H)** — using the **Sympy** library.

dcc/dt = 55.4872193927250, dcc/dh= 3.59430402988865

(The symbolic sensitivities represent *local linear rates of change* for the model trained on a particular data split and normalization. These values are **not fixed**)

These derivatives represent the rate of change of **cloud cover (CC)** in response to a unit change in each corresponding variable.

7.2 Interpretation of Results

- A **1-unit increase in temperature (T)** leads to an **approximate 55.48-unit change** in predicted cloud cover.
  This indicates a strong sensitivity of cloud cover to temperature variations, suggesting that thermal conditions play a crucial role in atmospheric condensation processes.
- A **1-unit increase in humidity (H)** results in an **approximately 3.59-unit change in the opposite direction**, reflecting an inverse sensitivity relationship under the dataset's normalized conditions.
  However, this sign direction depends on both the **data normalization** procedure and the **regression coefficient polarity**, meaning the exact physical interpretation must consider the original data scales.
- The relatively higher magnitude of ($|\partial CC/\partial H|$) compared to ($|\partial CC/\partial T|$) implies that **humidity** exerts a **stronger overall influence** on cloud cover fluctuations than temperature, consistent with meteorological expectations.

8. VISUALIZATION SUMMARY

8.1 Heat Map

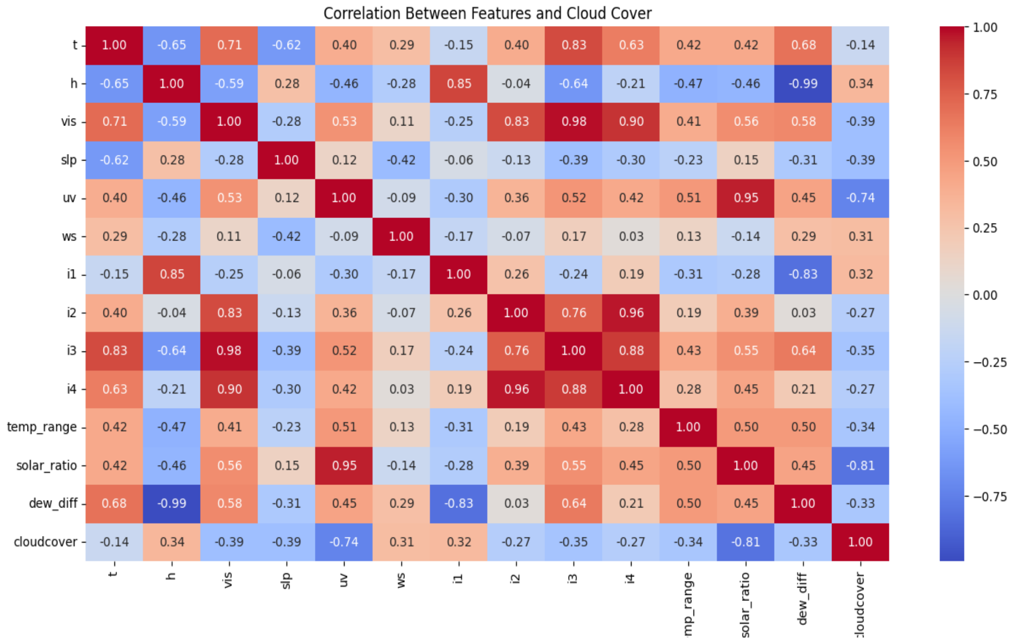

Figure: Correlation Heatmap showing linear relationships between meteorological features and cloud cover. Humidity exhibits a strong positive correlation with cloud cover, while visibility shows a moderate negative correlation.

8.2 Feature Importance (Linear Coefficients)

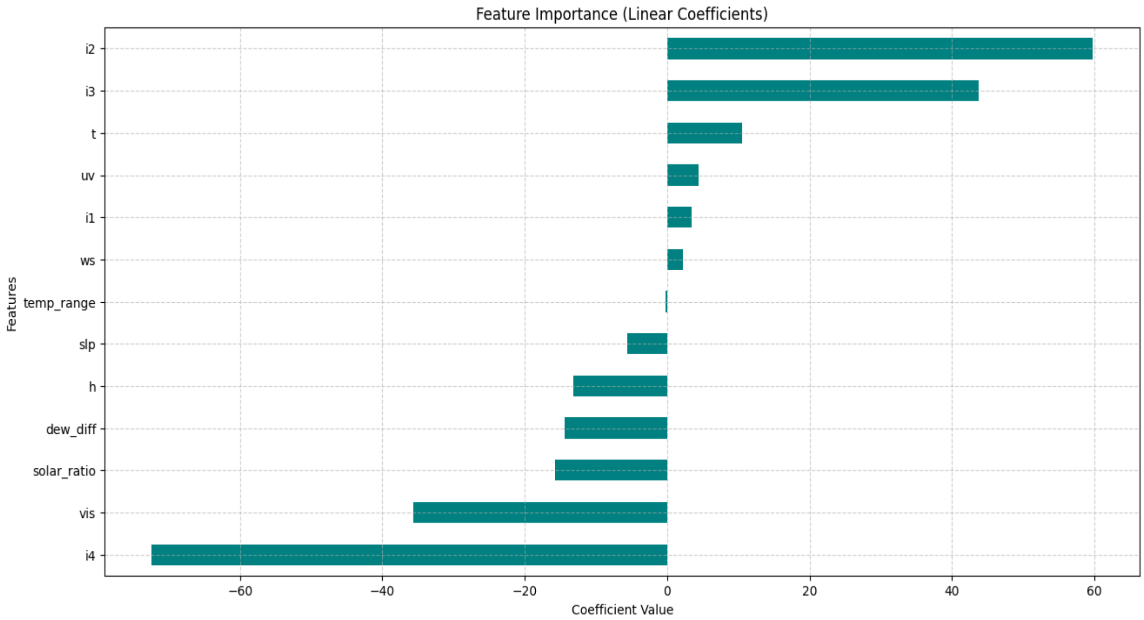

Figure: Feature importance plot based on linear regression coefficients. Humidity and temperature are identified as the most influential predictors of cloud cover, confirming their dominant role in atmospheric modeling.

8.3 Temporal Variation of Cloud Cover and Humidity

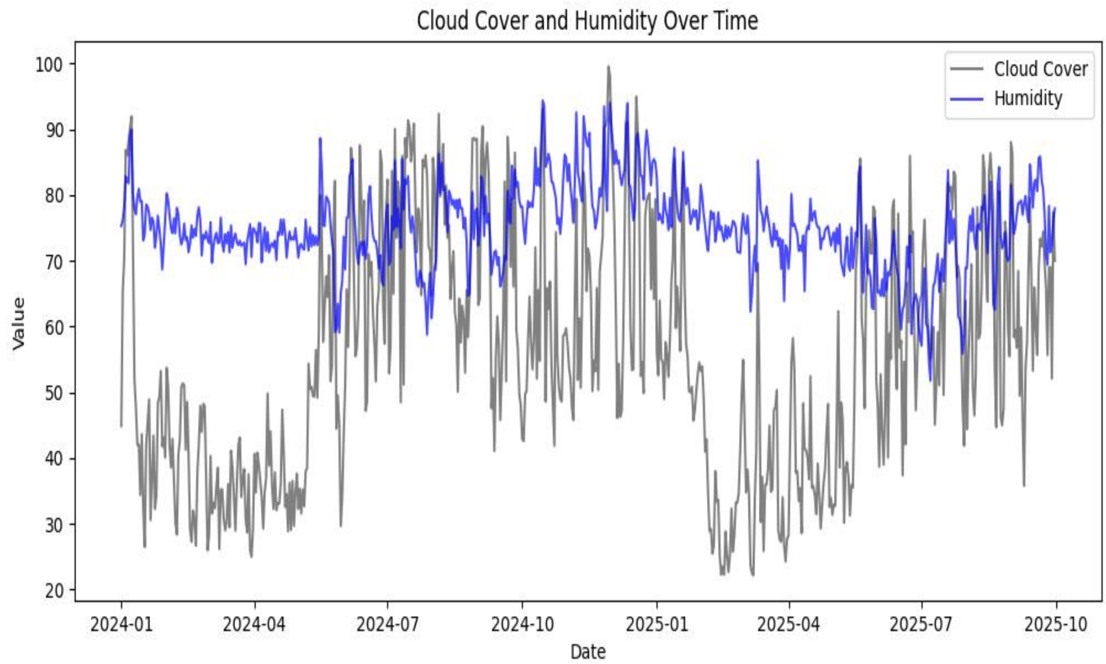

Figure: Time-series visualization of cloud cover and humidity over the observation period. Peaks in humidity correspond closely with increased cloud cover, reflecting consistent atmospheric moisture–cloud relationships.

several visual analyses were performed to explore data patterns and assess model performance. These included correlation heatmaps, feature importance plots, error distribution histograms, and time-series visualizations of key meteorological parameters. The visualizations provided valuable insights into the relationships between temperature, humidity, visibility, and cloud cover, as well as the model's predictive accuracy.

Note: Only a selection of representative charts is presented in this report for clarity and conciseness.

9. DISCUSSION

The results highlight that **humidity, visibility, and temperature** are the most significant factors influencing cloud cover variation in Chennai. Among these, humidity exhibited the strongest positive relationship, consistent with the physical understanding that higher moisture content favors cloud formation. Visibility, on the other hand, showed a negative correlation, suggesting clearer atmospheric conditions are typically associated with lower cloud presence.

Incorporating **interaction terms** (such as *temperature* × *humidity* and *temperature* × *visibility*) notably improved model stability and predictive accuracy. These terms allowed the regression model to partially capture **nonlinear dependencies** that simple linear relationships might overlook, enhancing the model's overall explanatory power.

Despite the model's good performance (average $R^2 \approx 0.70$), some **unexplained variance** remains. This may be attributed to **unmodeled atmospheric dynamics**, including vertical air movement, aerosol concentration, or local microclimatic variations that were not available in the dataset. Moreover, the use of surface-level measurements inherently limits the model's ability to capture higher-altitude meteorological processes influencing cloud formation.

Overall, the findings validate that surface meteorological parameters can be effectively used for **computationally efficient cloud cover estimation**, although inclusion of additional physical parameters or advanced nonlinear models could further improve prediction reliability.

10. CONCLUSION

This study demonstrated that cloud cover over Chennai can be effectively estimated using a multiple linear regression framework built entirely with NumPy. By analyzing historical surface-level meteorological data, the model identified **humidity, temperature, and visibility** as the dominant predictors of cloud cover. The inclusion of **interaction terms** improved the model's ability to capture subtle nonlinear effects, leading to consistent predictive performance across different dataset partitions.

Exploratory Data Analysis (EDA) and visualization confirmed meaningful physical relationships between variables, while sensitivity analysis provided quantitative insight into how small variations in temperature and humidity influence cloud cover levels. Although the model achieved an average **R^2 score of around 0.70**, results varied slightly between runs due to differing train-test splits and normalization factors.

Overall, the regression-based approach offers a **computationally efficient and interpretable alternative** to satellite-based or physically intensive forecasting models. Future work could focus on extending the dataset to include **additional atmospheric parameters** (e.g., wind shear, pressure gradients, aerosol index) or integrating **nonlinear and machine-learning techniques** to further improve accuracy and generalization.

11.REFERENCE:

- The Relationship between Cloud Cover to Temperature and Humidity – Llyn L. Escarcha And Ashanti F.Pangue.
- NumPy Developers, "numpy.linalg.lstsq — Least-squares solution to a linear matrix equation," *NumPy Documentation*, 2025.
- Matplotlib Developers, "Matplotlib: Visualization with Python (Version 3.x)," *Official Documentation*, 2025.
- An Introduction to Statistical Learning with Application in R – Gareth James, Daniela Witten.